

Spring Cleaning 102:

File organization strategies

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Why “Spring Cleaning” for Research Data

Research Data Accumulates Fast

- A new project begins while another one is ongoing
- Your work grows and brings in multiple collaborators (internal and external)
- Your data gets spread across multiple storage locations.

Little time to stop and tidy

The Cost of Not Cleaning Up

- You risk losing or duplicating your data
- Researcher turnover can cause confusion about versions.
- An attempt to migrate can be painful.



Today we will cover



Why good data management matters



Core practices for managing research data



FAIR Principles in concept



Data Storage and Management Strategies



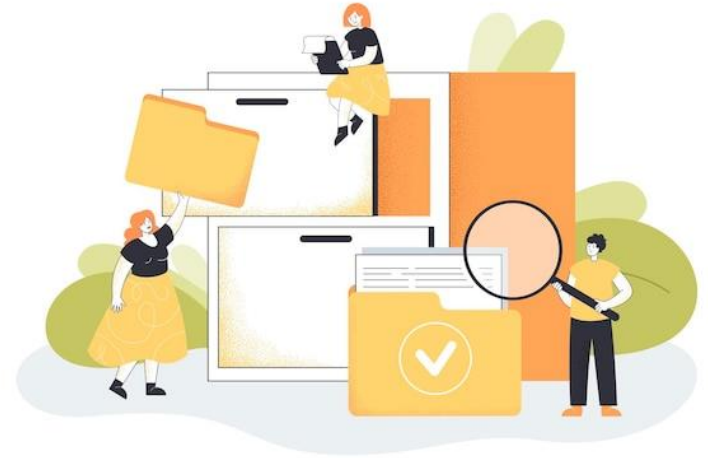
Data Movement and Migration

What is “Good” Data Management?

At its core, Organized Data is:

- Easy to **navigate**
- Easy to **explain**
- Easy to **move**
- Easy to **reuse**

*File organization is the **foundation** of every other data management practice.*



ABC of Data Management

- **Arrange**
 - your data into meaningful collections.
- **Backup**
 - your data so you do not lose anything important.
- **Classify**
 - your data with labels and categories for quick access.



Alejandro, D. (2024). *Data Management for Kids*.

Backups are more than just copies of your data

Backups:

- are systematic and purposeful
- preserve version history
- are designed for recovery scenarios
- include integrity checks
- are stored separately and securely
- can be automated to reduce human error
- support compliance and reproducibility
- can be centralized and managed



Why Good File Organization Helps

Immediate Benefits

- Get data quickly to your compute source
- You reduce errors in data searchability
- Makes collaborating with your colleagues easier
- Reduces stress when deadlines hit

Long-Term Benefits

- Your data is reproducible
- You can publish with confidence
- Helps maintain compliance
- Improves reusability by you and others



Recap: Storage Options at Northwestern

Storage	Who is it for	Used For	Key Feature
OneDrive	Individual/External	Storing working data that only you need to access	Auto-sync, version history
SharePoint	Group/External	Storing working data shared with a team	Controlled sharing
RDSS (resfiles, resfiles-aduit, fsmresfiles)	Group	Storing working data shared with a team, especially data with large individual file sizes	Reliable, centralized
Public Cloud (Google Cloud, AWS, Azure)	Group/External	Storing working data and archiving research data	Easy collaboration, pay-per-use
Quest (High-Performance Computing)	Group	Storing data being actively analyzed on Quest	Secure, large-scale storage
Computers (Lab, Office, Instruments, Core Facility)	Individual/Group	Storing data that you and your team members require	Supports legacy systems

Storage used by Researchers

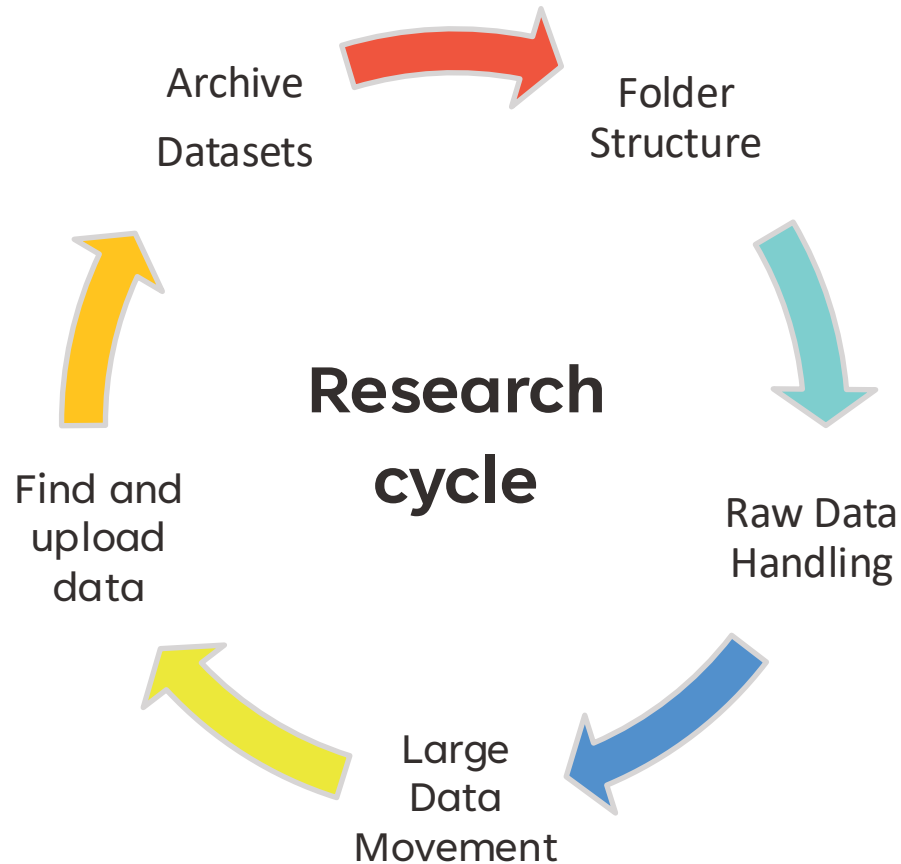
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Self Managed Storage (Hard Drives, NAS servers, Dropbox, Box, Google Drive)	Individual/Group/External	All the above	Like the other options, but without the support

Where do you start?

File Organization Across the Research Lifecycle

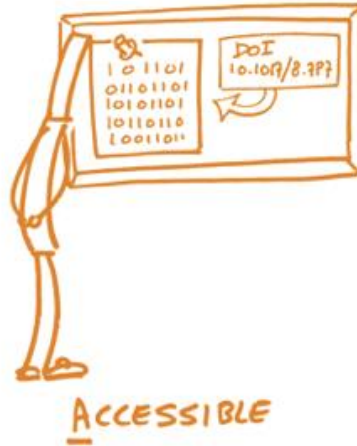


File Organization Across the Research Lifecycle



FAIR

FAIR DATA PRINCIPLES



- Logical, consistent folder structures
- Clear, descriptive file and folder names
- Clear permissions and access rules
- Authorized users know **how** to retrieve the data
- Folder structures that don't rely on personal knowledge
- Metadata stored **with** the data
- Raw and processed data clearly separated
- Documentation explains context and methods

*Načela FAIR - DiRROS Data. (2025, April 18). DiRROS Data.
<https://dirrosdata.ctlk.uni-lj.si/en/raziskovalni-podatki/nacela-fair/>*

Ask the right questions



Can a new member in your project easily **find** and understand your data?



If you stepped away for six months, can you **access** your data and make sense of it again?



Can your files and formats **interact** smoothly with other systems or collaborators?



Could someone else **reuse** your data, or even you, years from now?

One Size Does Not Fit All

Key Considerations



Type of Research



Type of Data (File size, type, volume, velocity, and variety)

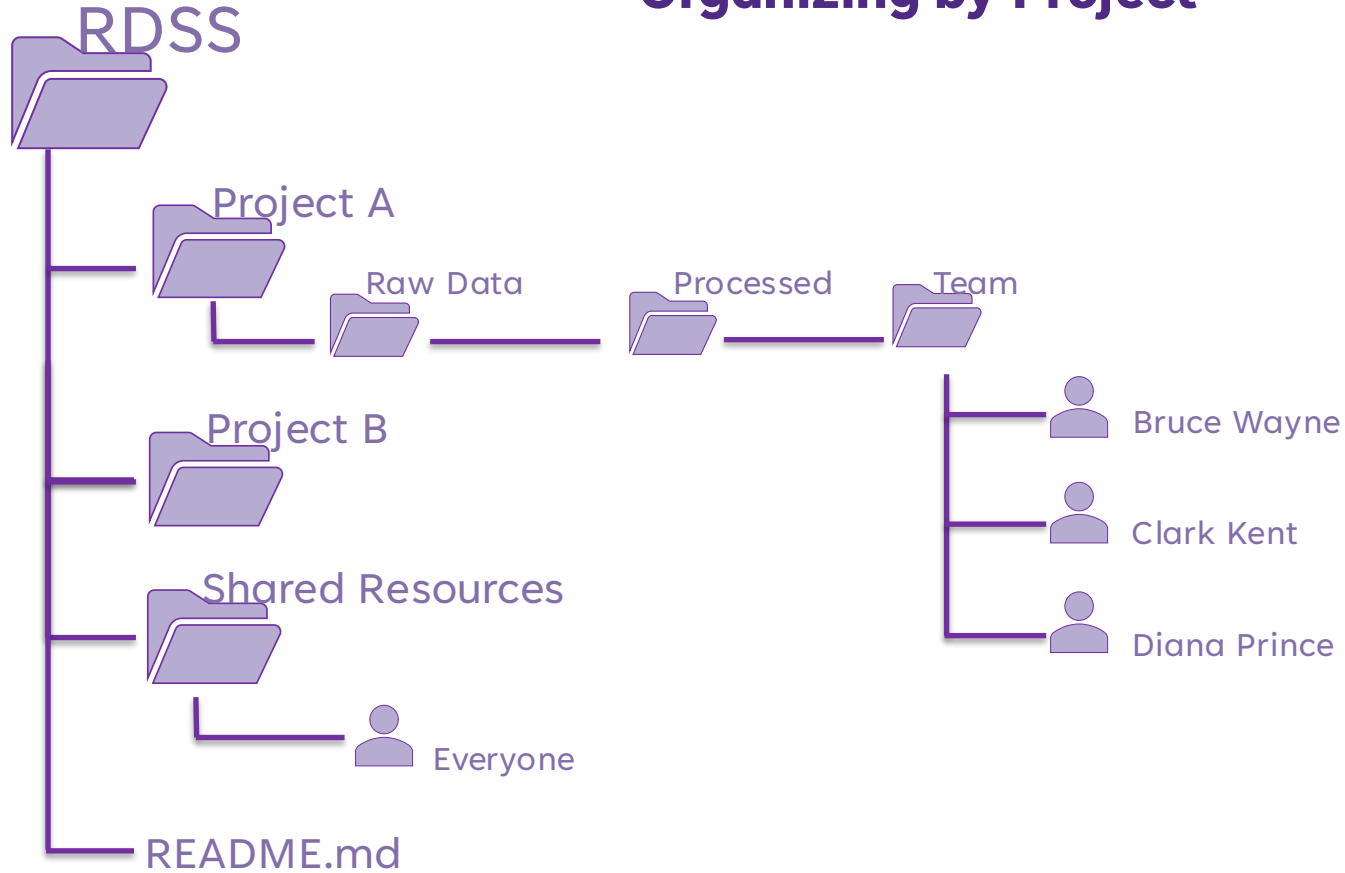


Sensitivity

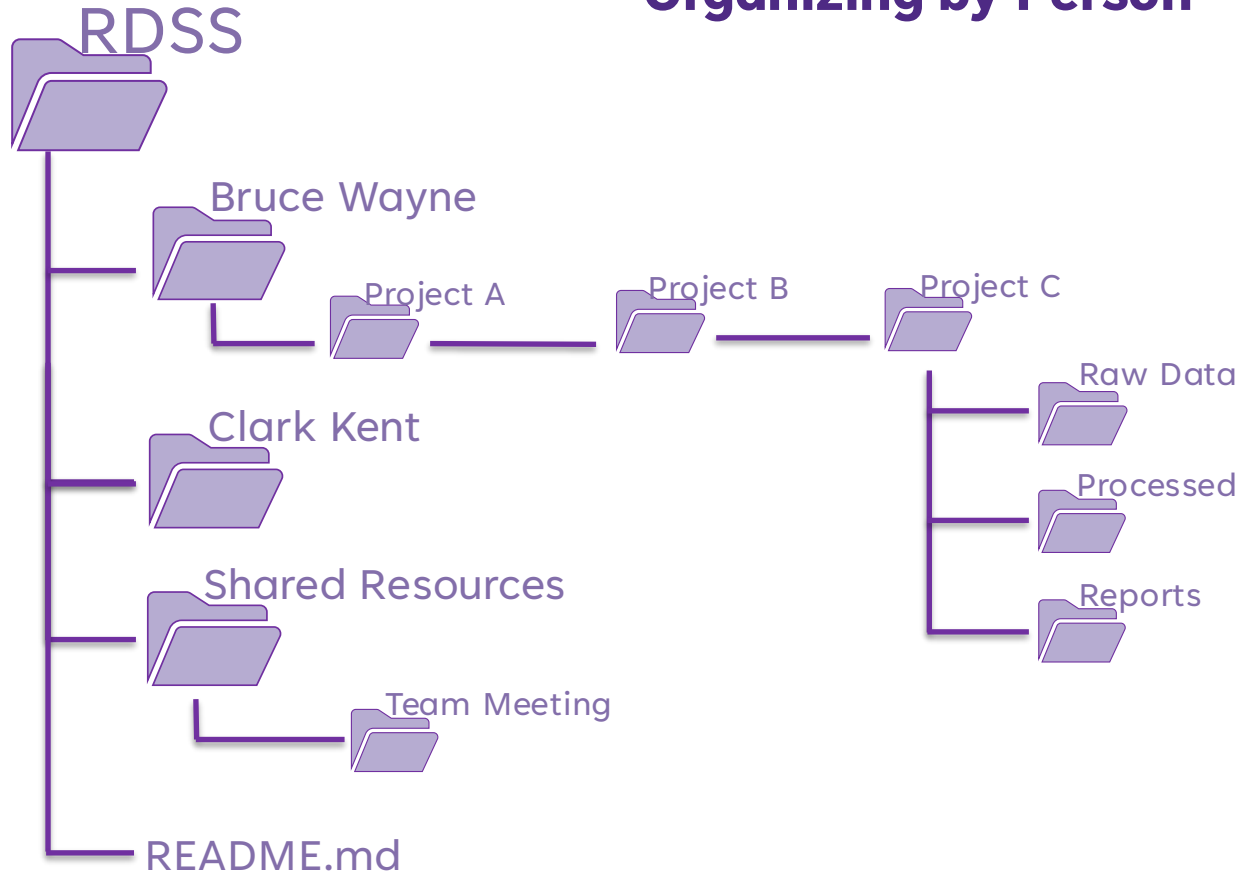


Shareability

Organizing by Project



Organizing by Person



Deep Folder structure challenges

No technical limit to how deep a folder structure can go, but...

Harder to Navigate

- People get lost in long, nested paths.

File Path Length Limits

- Some systems (especially Windows and Globus) **reject long paths**, causing errors in saving, syncing, or backups.

Redundant Structure

- Too many levels often duplicate information already in filenames

Collaboration Friction

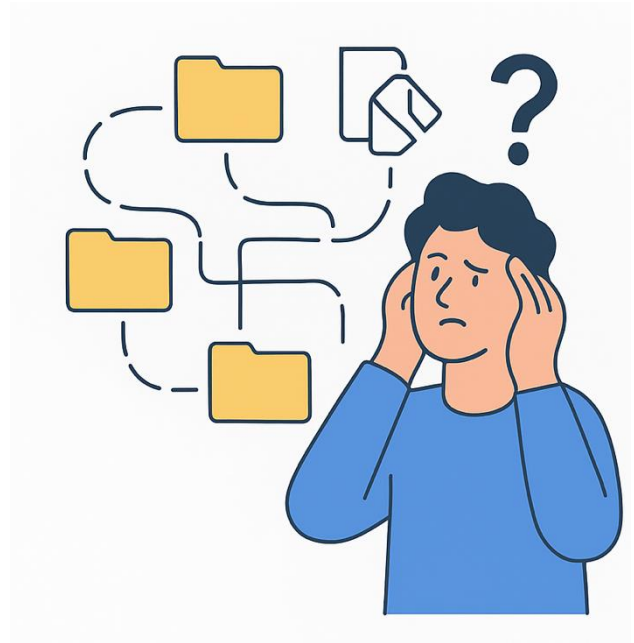
- Teammates may not know *where* to store or find files.

Transfer and Archival Issues

- Long paths and nested folders can cause **missed files** during automated backup or archiving processes.

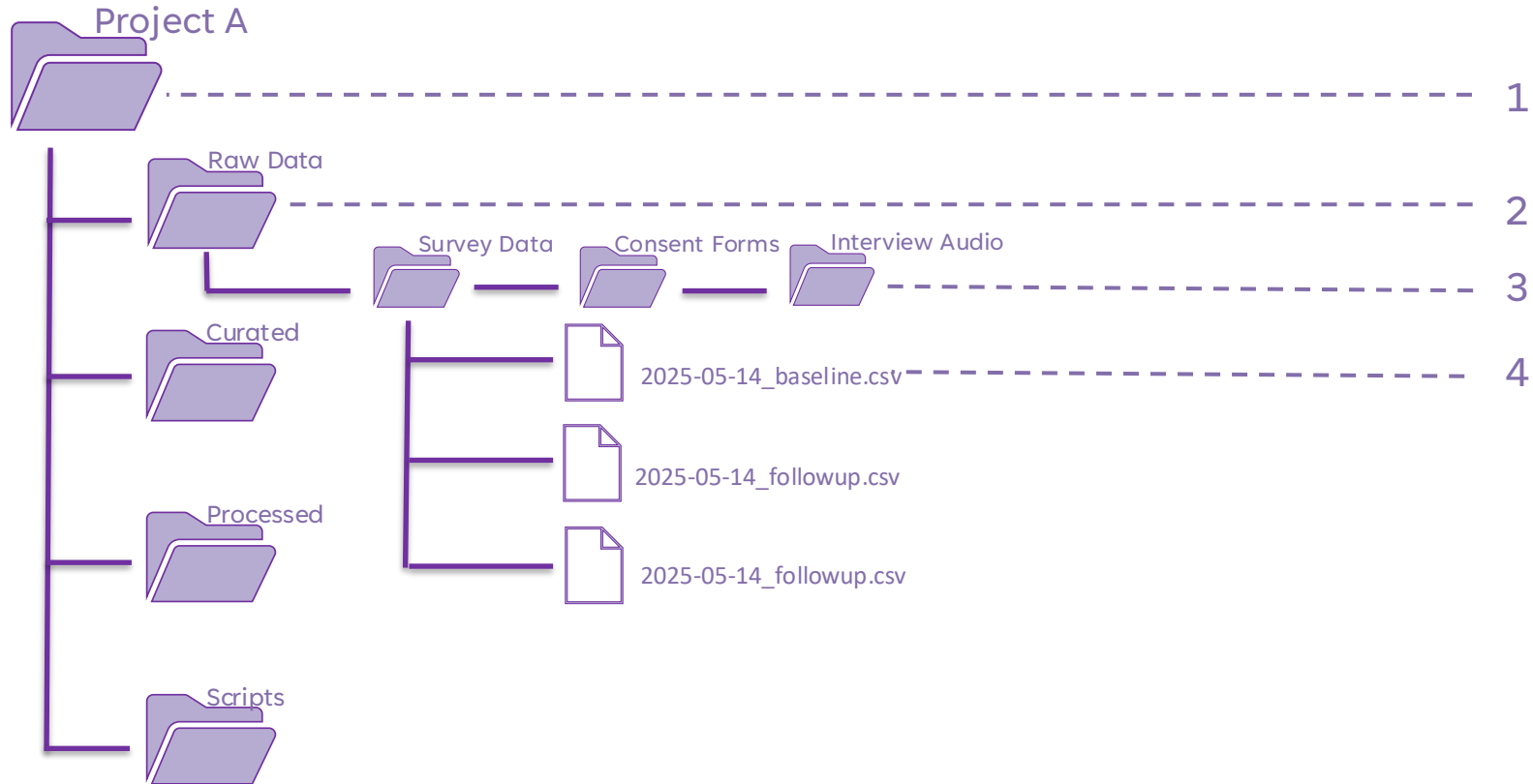
Cognitive Load

- The more levels, the harder it is to mentally map the structure



Follow the 4-level Rule...

The 4 Level Rule



Make it work for your Research

Genomics & Bioinformatics

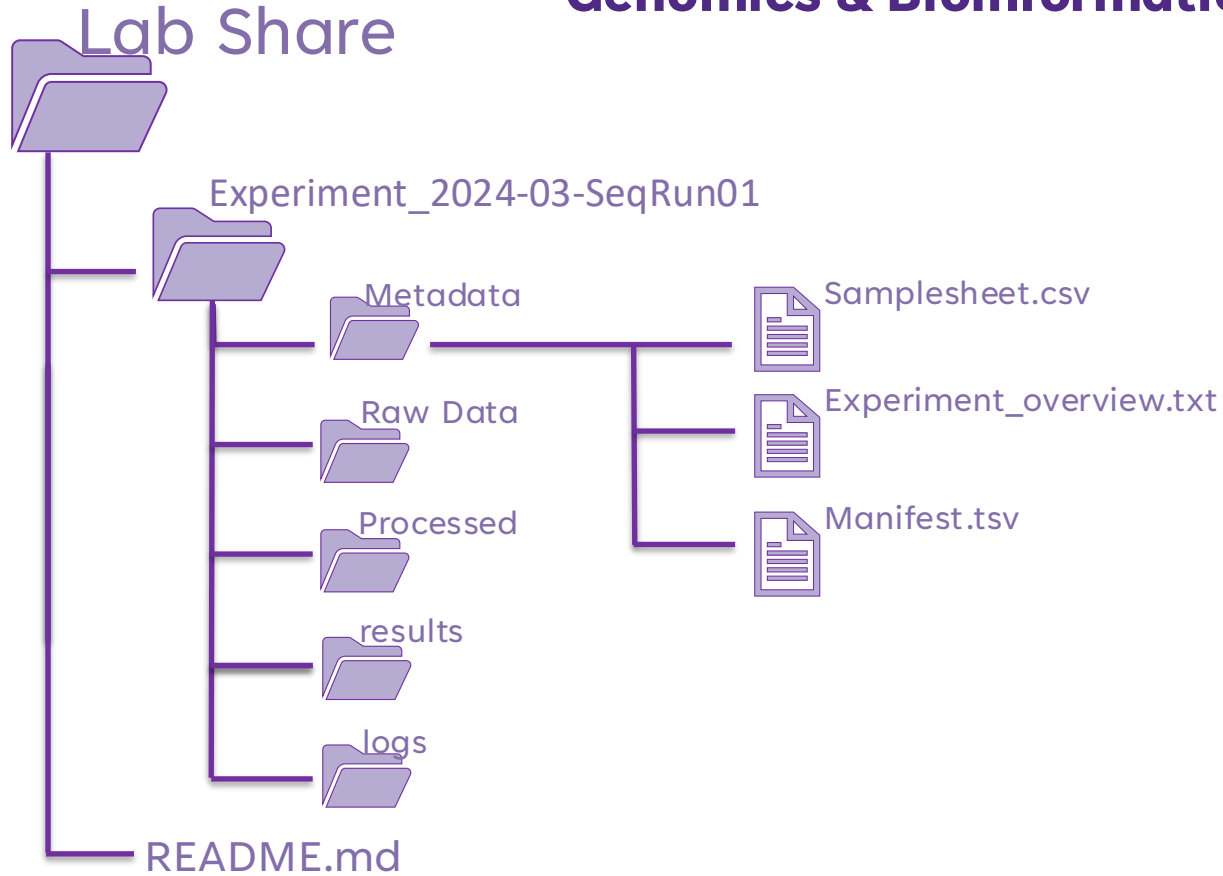
File & Storage Strategy

- Instrument PCs: **temporary capture only**
- On-prem or secure cloud: **raw data storage**
- Quest: **analysis and intermediate outputs**
- Clean separation of:
 - Raw data
 - Intermediate files
 - Results and summaries
- Metadata and manifests are stored with each dataset

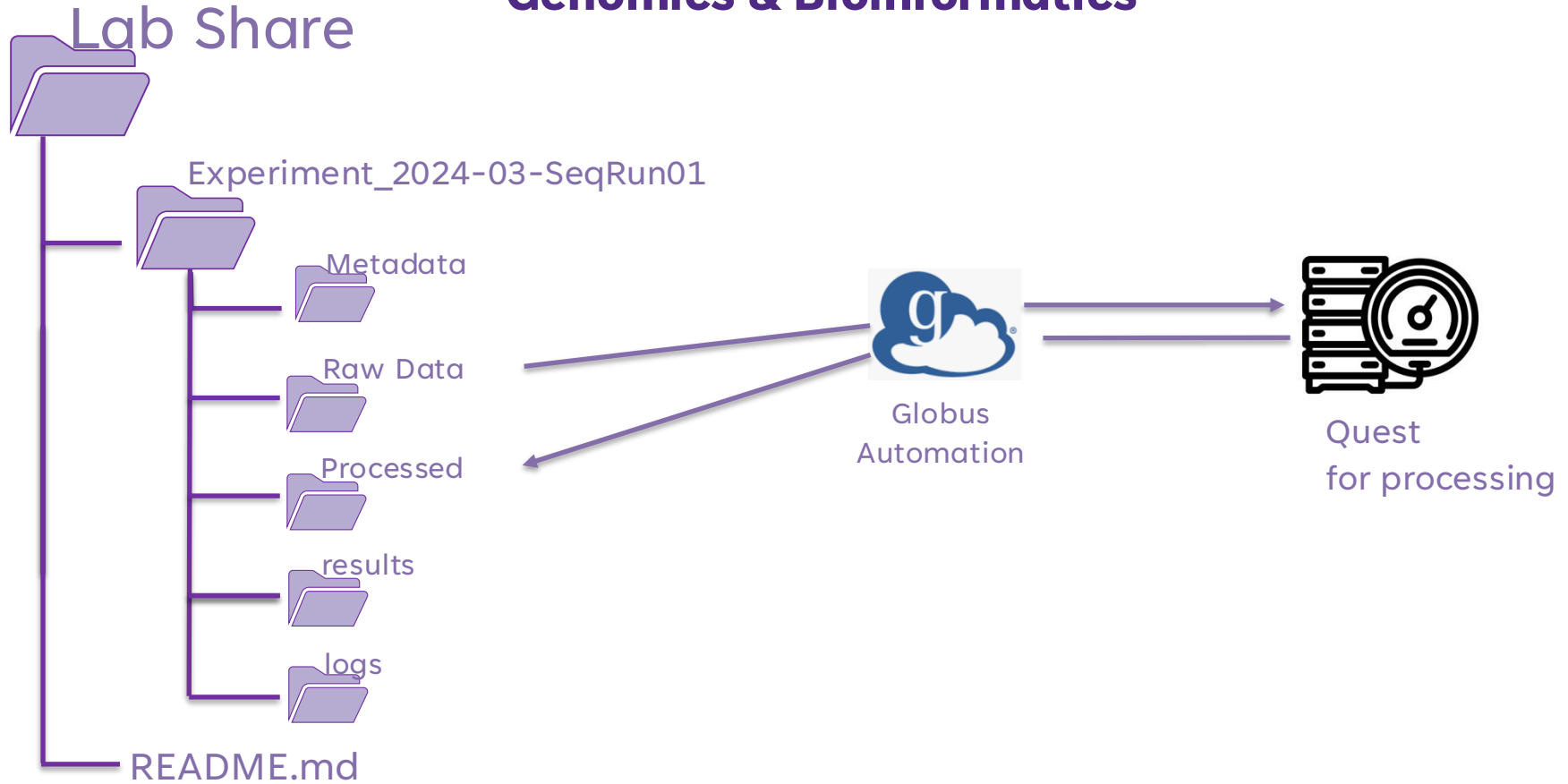
How Globus Helps

- **Automatically** move large sequencing files off instruments
- **Reliably transfer data** between on-prem storage and HPC
- **Share curated results** with collaborators via shared collections
- Conduct **repeated**, large-scale transfers without manual effort

Genomics & Bioinformatics



Genomics & Bioinformatics



Health & Clinical Research

File & Storage Strategy

- Instrument or collection systems:
short-term staging
- Secure on-prem or compliant cloud:
primary working data
- Clear separation of:
 - Identifiable sensitive data
 - De-identified analysis data
- Documentation and data dictionaries are stored alongside datasets

How Globus Helps

- Move data securely without email, USB drives, or ad-hoc tools
- Control access for internal and external collaborators using shared collections
- Automate transfers between secure systems
- Maintain auditability and trust during data movement

Social Sciences

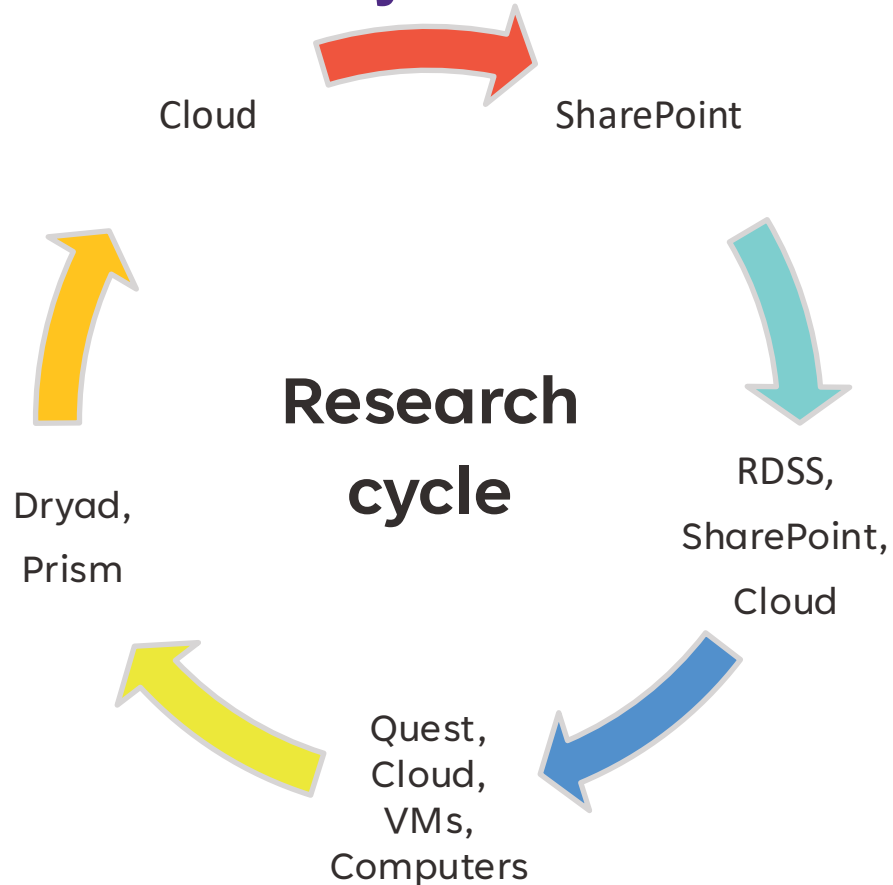
File & Storage Strategy

- On-prem or cloud storage for active work
- Clear separation of:
 - Raw data
 - Cleaned datasets
 - Analysis scripts
 - Published outputs
- Strong emphasis on documentation and version clarity

How Globus Helps

- Easily move cleaned or final datasets to repositories
- Share restricted datasets with collaborators without duplication
- Preserve consistent structure across storage systems
- Simplify handoff from active research to archiving or publication

Where should your data be stored?

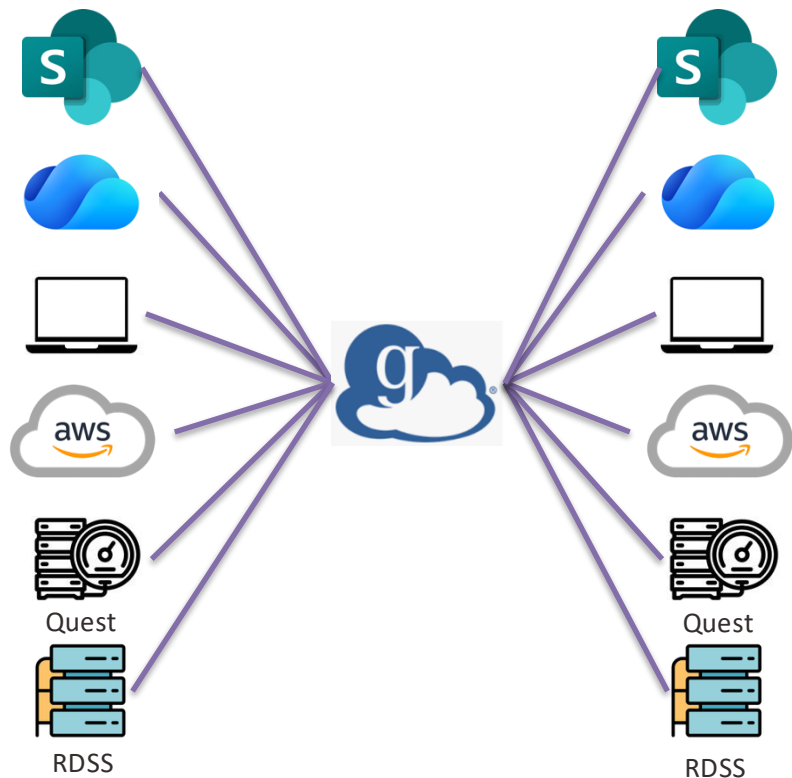


Moving or Migrating Your Data

Work with Globus

Globus ensures:

- Your **organized structure is preserved**
- Large transfers **don't fail halfway**
- Data integrity **can be trusted**
- Transfers complete **without manual monitoring**
- Partial or corrupted copies are **avoided**
- Transfer history provides **traceability and auditability**
- Data arrives **exactly where workflows expect it**
- Automation supports **repeatable, lifecycle-based movement**



Globus is our recommended method for data transfer

Globus can help

Well-structured folders make management clear

- Use Globus to manage what data needs to be moved
- Organize what data stays put
- Identify what stage the data is in

Create shared / guest collections with controlled access

- Collaborate with **external researchers** safely
- No data duplication, no email attachments
- Globus controls support sensitive and regulated data

Set up automations to move or back up data between

- Instrument PCs to On-prem / Quest
- Quest to Cloud / On-prem when analysis finishes
- Final outputs to archive or repository

Build repeatable pipelines for

- Ingest → process → clean → analyze → archive
- Reduce manual errors and forgotten files
- Track and label transfers for context and provenance

Finally

Good Organization Practices

There is no one right answer
Formulate a plan
Be consistent



Portability: Put all project files in one folder



Findability: Use descriptive file names



Navigability: Flatten your subfolder structure



Integrity: Use controlled access and backups to protect your data

Naming Best Practices



Consistency is key: Use naming conventions that help the future you



Standardize: Add in standards that help your system find/query your data efficiently



Descriptive yet meaningful: Find the right balance between informative without overload



Reproducibility: Document your approach, work with your team

If you plan, you're ahead of the game

Structure with purpose

Organize around projects or people. Keep folder depth manageable (≤ 4 levels).

Design File Names with Intent

Use clear, consistent, and compact naming conventions; your future self will thank you.

Document everything

A simple README beats hours of detective work later. Explain structure, naming, and abbreviations.

Automate where possible

Let Globus handle backups and do the heavy lifting with transfers

Plan for portability

Well-organized data moves easily — whether you're changing platforms, collaborators, or roles.



FIND WHAT YOU NEED



PLANNING

- [Writing a Data Management Plan](#)
- [Protecting the Sensitive Information in My Data](#)



DATA COLLECTION AND STORAGE

- [Choosing Appropriate Storage](#)
- [Documenting Your Research](#)
- [Transferring Data to or from Northwestern](#)
- [Sharing Data with an External Collaborator](#)



DATA SHARING AND ARCHIVING

- [Making Your Data Reusable](#)
- [Sharing Data Publicly](#)
- [Archiving Data When a Project is Done](#)



SUPPORT AND RESOURCES

- [Talk to a Data Management Expert](#)
- [Northwestern Research Data Management Resources](#)
- [External Research Data Management Resources](#)

Reach out!

Visit the: [Research Data Management Website](#)

Email: researchdata@northwestern.edu

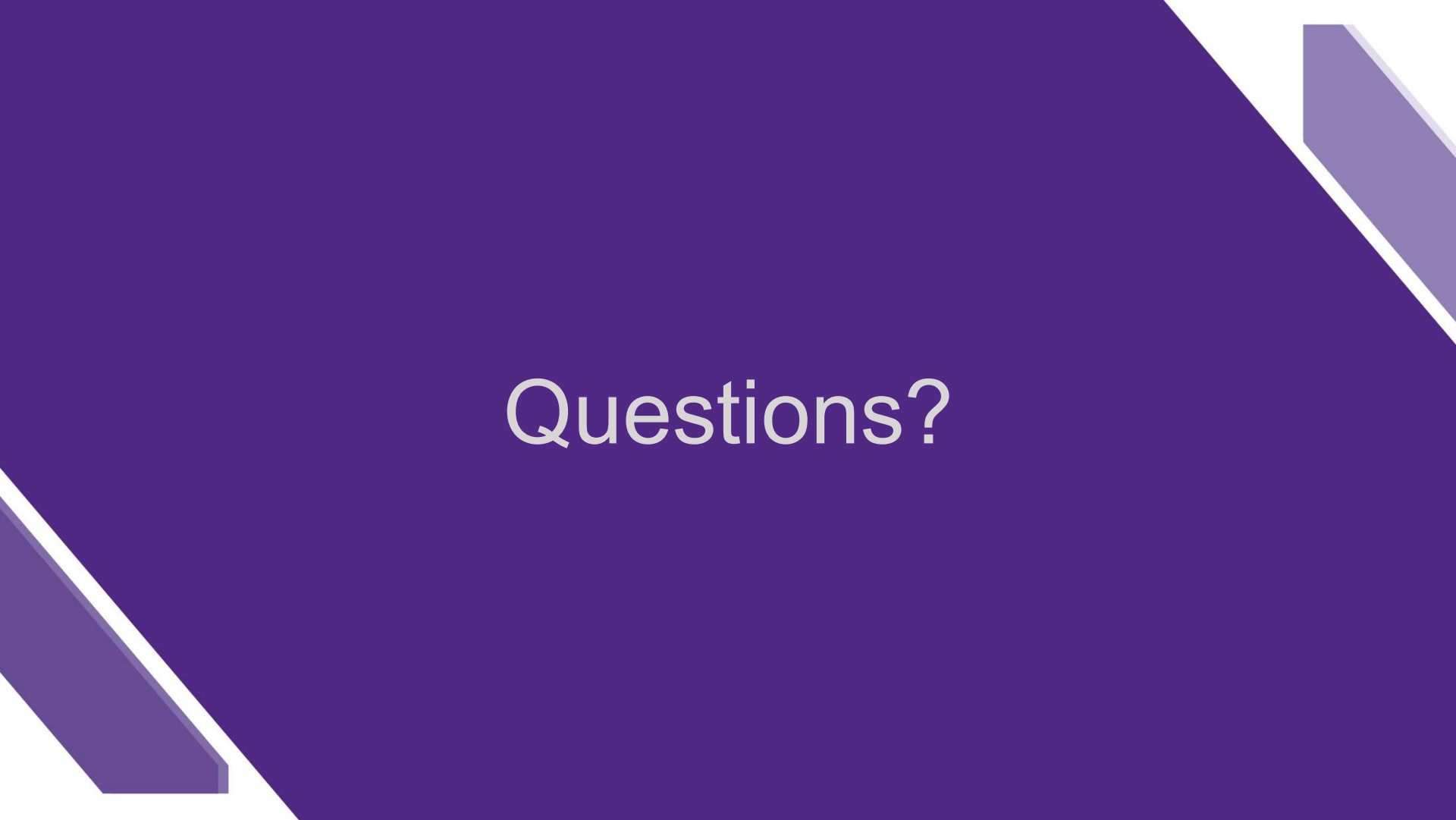
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Questions?